
REPORT OF ANALYSIS - PHYSICS, PARTICLE & SURFACE ANALYSIS

Sample:	TB1175B
Order title/Product:	Graphene / Raman microscopy
Project:	Graphene
Account:	Curia Germany GmbH
Customer:	Dr. Thorsten Busch
Order no.:	25-050027-1
Remarks:	Raman measurement on request of the customer

Method:

Raman spectroscopy/microscopy using Senterra I Raman microscope (Bruker).

Procedure:

Objective:	Raman spectroscopy
Field of Application:	Powder

Principle:	The substance is applied to a quartz microscope slide and examined under the microscope. The objective is selected and focused to examine an exact location using Raman. At this position, several points are examined with the laser and a corresponding number of Raman spectra are recorded.
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Reference:	N/A
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Experimental Details and Result

Equipment:	Senterra I Raman microscope (Bruker)
Magnification:	20 x
Wavenumber range:	45 – 3700 cm ⁻¹
Wavenumber step size	3 – 5 cm ⁻¹
Co-addition:	100
Exposure time:	10000 ms
Evaluation:	Averaging over similar spectra

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Result:

Absorption band	2D band	G band	D band	D/G ratio
Integral _{Typ A}	2,3	217,1	290,3	1,337
Integration range	2796 - 2537	1743 - 1468	1468 - 954	
Peak high	0,021	1,948	1,637	

Spectra see attachment.

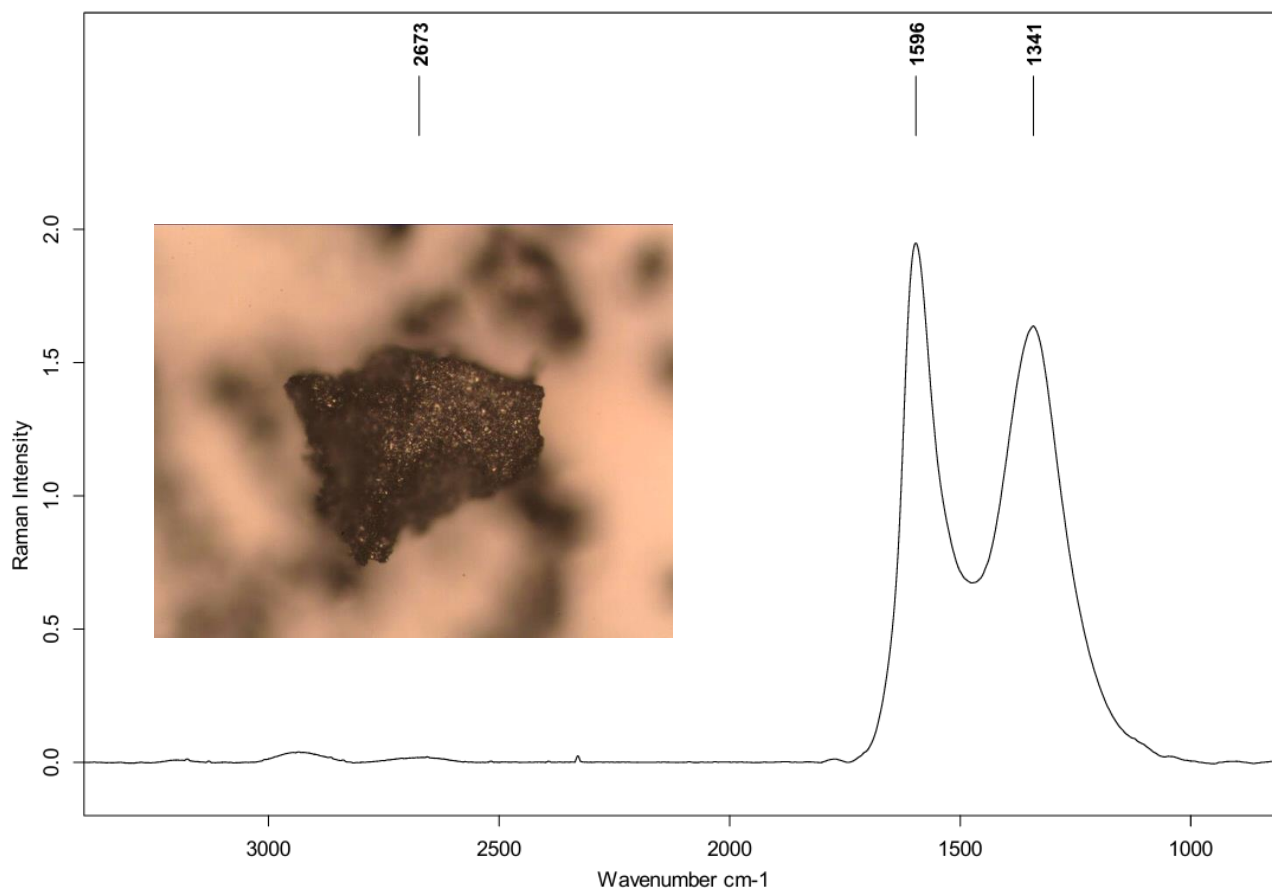


Figure 1: TB1175B, average spectrum over 8 recordings

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